A Calibrated Transcriptomic Atlas of Porcine Development Reveals Conserved Molecular Programs with Humans

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Abstract

Experimental reproducibility and cross-species translation using the domestic pig (*Sus scrofa*) are often limited by the lack of a standardised molecular framework to define biological maturation. To address this, we developed a calibrated transcriptomic atlas that maps porcine development across five major tissues, providing an objective standard for cross-species synchronization. By applying machine learning to 1,924 RNA-seq profiles, our framework achieves high-precision staging (mean accuracy: 0.83) and reveals that distinct organs mature via highly tissue-specific molecular programs. Crucially, we validate this atlas as a bridge for human research through a direct comparison with human skeletal muscle development, finding 89% directional concordance in gene trajectories and a significant correlation ($r = 0.62$) in conserved regulatory modules, including *SORCS2* and *FGFRL1*. These results demonstrate that porcine molecular maturation biologically mirrors the human program, providing a rigorous, scalable standard that enables researchers to more accurately perform cross-species validation studies in nutrition, physiology, and disease modelling by precisely matching porcine subjects to human developmental stages.

1 Introduction

The domestic pig (*Sus scrofa*) is central to modern animal science, serving as both a major agricultural species and an increasingly important model organism[1–3]. However, a critical gap undermines reproducibility across pig research: the lack of a standardised, molecular definition of developmental stage. Current staging relies on morphological criteria or chronological age, which are poor proxies for biological maturity[4]. Littermates frequently develop asynchronously, and distinct organs mature at different rates, effectively decoupling whole-body age from tissue-specific functional competence. This imprecision confounds experimental reproducibility in nutrition, growth physiology, and disease studies, where developmental stage is a critical variable[5].

Molecular approaches offer a solution to this challenge. Epigenetic clocks, which leverage DNA methylation patterns to predict biological age, have emerged as powerful tools for quantifying organismal maturation[6–9]. Similarly, transcriptomic profiles provide a quantitative readout of cellular state, enabling precise biological staging that transcends morphological variation[10,11]. While such molecular clocks are established in humans and mice[12,13], a comprehensive, tissue-resolved developmental atlas has been missing for the pig, limiting the standardisation of experimental protocols across research groups.

Here, we present the first calibrated transcriptomic atlas of porcine development, integrating data across five major tissues (brain, liver, muscle, lung, and blood) with sufficient sample sizes for machine learning analysis. Unlike previous descriptive studies, we build a quantitative, predictive framework that infers developmental stage with high accuracy directly from gene expression profiles. To validate the biological relevance of our atlas, we performed cross-species comparison with human skeletal muscle, revealing striking conservation of developmental trajectories (Pearson $r = 0.62$) and identifying shared regulatory modules that orchestrate postnatal maturation in both species. We note that cross-species validation is currently limited to muscle tissue due to availability of matched human developmental data. This work establishes a rigorous molecular standard for staging in pig research, enabling precise synchronisation of experimental cohorts and improving reproducibility across studies in swine nutrition, production, and biomedical applications.

2 Results

2.1 A comprehensive transcriptomic atlas of porcine development

To define a molecular standard for porcine development, we constructed a large-scale reference atlas using RNA-seq profiles from the PigGTEx consortium[14–17] (Fig. 1a). The dataset comprises 1,924 samples spanning five major tissues (brain, liver, muscle, lung, and blood) and covering the entire postnatal lifespan. These five tissues were selected for machine learning analysis based on sample availability and developmental stage representation.

We addressed the limitations of chronological age by defining five calibrated life stages based on physiological milestones: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d) (Fig. 1c). These stages correspond to distinct physiological transitions: weaning completion ($\sim$20 d), early growth phase ($\sim$60 d), puberty onset ($\sim$150 d), and reproductive maturity ($\sim$365 d)[1]. This provided a ground truth framework for training, allowing us to decouple "biological time" from simple chronological time. The resulting dataset serves as the most comprehensive transcriptomic resource for porcine development to date.

2.2 Accurate inference of physiological developmental stage

We implemented a calibrated machine learning framework to infer the biological developmental stage directly from tissue-specific transcriptomes (Fig. 1b). Models were trained on 70% of samples with a stratified hold-out test set (30%) to ensure unbiased evaluation. Test set performance demonstrated robust generalisation across tissues, with a mean balanced accuracy of 82.5% (95% confidence interval [CI]: 79.2–85.8%) across all tissues (Fig. 2a; see Supplementary Table S1 for detailed metrics).

Performance varied according to tissue complexity and sample availability. The liver, a highly homogeneous metabolic organ with balanced class representation, achieved the highest balanced accuracy (92.0%, 95% CI: 88.5–95.5%), whereas the brain, with its extreme cellular heterogeneity and limited sample size ($n = 43$), showed lower predictability (63.8%, 95% CI: 49.2–78.4%). Notably, tissues with severe class imbalance required reduced classification schemes (4-class for muscle and liver, 2-class for brain, blood, and lung; Fig. 1d) to ensure robust model training (Supplementary Table S1).

Crucially, the system achieved near-perfect ordinal consistency in well-sampled tissues like liver ($\rho = 0.98$) and muscle ($\rho = 0.94$) (Fig. 2c). The confusion matrices (Fig. 2b) reveal that "errors" largely occurred between adjacent stages (e.g., Early vs. Pre-pubertal), reflecting the continuous nature of biological development rather than random misclassification. Unlike static morphological grading, our probabilistic calibration captures these transitional states.

These tissue-specific performance differences prompted us to ask: what biological programs underlie the model's predictions? To answer this, we examined the functional annotation of the genes that drive accurate stage inference.

2.3 The functional landscape of development

Beyond prediction, our model uncovers the biological logic driving maturation. We performed functional enrichment analysis on the top informative genes for each stage, mapping the global landscape of developmental regulation (Fig. 3).

The resulting functional network reveals that each tissue follows a distinct regulatory framework. In the brain, development is driven by synaptic refinement and metabolic support[18,19], highlighted by enrichment of *sarcosine metabolism* (linked to NMDA receptor coagonist availability[20]) and *beta-tubulin binding* (critical for axonal transport). In contrast, the liver prioritises metabolic autonomy and homeostasis, evidenced by the upregulation of *autophagosome-membrane adaptors*. Muscle development centres on contractile machinery assembly and neuromuscular signalling (e.g., *acetylcholinesterase activity*[21]), while the lung focuses on immune system establishment[22] (*IL-6 signalling regulation*) and RNA metabolic remodelling (*m6A reader activity*). This systems-level view moves beyond simple marker genes to capture the coordinated, tissue-specific logic of physiological maturation.

Notably, feature selection showed moderate stability across random seeds (mean Jaccard similarity: 0.16–0.43 across tissues; Supplementary Fig. S2b), with 10–60% of top features appearing in $\geq$80% of runs. This variability reflects biological gene redundancy rather than model instability. The selected features showed strong correlations with developmental stage (mean absolute $\rho = 0.4$–0.7; Supplementary Fig. S2c). Critically, cross-tissue feature overlap was extremely low (0.3% Jaccard similarity). This profound tissue-specificity confirms that each tissue uses unique molecular programs for developmental progression.

2.4 Evolutionary conservation with human development

A central question for the pig model is whether its developmental timing biologically mirrors that of humans. We addressed this by performing a direct, quantitative comparison with human skeletal muscle transcriptomes from infant to adult stages[23,24] (Fig. 4). We note that this cross-species validation is limited to muscle tissue due to availability of matched human developmental data.

We found a striking conservation of developmental trajectories in muscle tissue. Using an adaptive thresholding approach to optimise signal-to-noise ratio, we identified 36 genes meeting significance criteria ($p < 0.10$) in both species. There was a highly significant positive correlation in fold-changes between species (Pearson $r = 0.62$, $p = 5.86 \times 10^{-5}$; Supplementary Fig. S1), with 89% (32/36) of shared developmental markers changing in the same direction (Fig. 4a). Bootstrap analysis (1,000 iterations) confirmed the robustness of this correlation (95% CI: 0.36–0.81; Supplementary Fig. S1b), and sensitivity analysis across p-value thresholds (0.05–0.20) showed consistent positive correlations ($r > 0.55$) when 15–50 genes were included (Supplementary Fig. S1a).

We identified three conserved functional modules driving this maturation (Fig. 4b,c; Supplementary Table S3).

Module 1: Neuromuscular Maturation. This module comprises genes with high infant expression that decreases during postnatal development. *ACHE* (pig FC = -1.81, human FC = -0.53), essential for neuromuscular junction function[21], shows elevated expression in the Infant stage (0–20 d; mean TPM = 16.2) that progressively declines through Early childhood (21–59 d; TPM = 9.9), Pre-pubertal (60–149 d; TPM = 9.1), to baseline in Post-pubertal stages (150–365 d; TPM = 1.3). *KREMEN1* (pig FC = -1.25, human FC = -1.43), a negative regulator of Wnt/β-catenin signalling that controls cellular proliferation[25,26], shows a similar gradual postnatal decline. *SORCS2* (pig FC = -1.84, human FC = -0.95), a regulator of motor neuron innervation[27], is highly expressed embryonically and during early postnatal life (Infant TPM = 14.6) and becomes progressively downregulated through Early childhood (TPM = 10.9), Pre-pubertal (TPM = 6.5), reaching its lowest in Post-pubertal (TPM = 0.6).

Module 2: Structural Refinement. This module reflects the transition from rapid tissue growth to structural consolidation. *FGFRL1* (pig FC = -1.68, human FC = -2.10) acts as a decoy receptor dampening excessive FGF signalling[28]; its expression peaks during the Infant stage (TPM = 71.2), drops sharply in Early childhood (TPM = 20.6), and decreases substantially to its lowest in Post-pubertal (TPM = 4.3). *DLK1* (pig FC = -1.37, human FC = -2.88), a crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation[? ?], shows high infant expression that declines substantially during postnatal development as the muscle fiber population matures. *PDLIM3* (pig FC = 2.19, human FC = 0.56), a structural Z-disc protein that stabilises actin filaments[30], shows the opposite pattern—lower infant expression (TPM = 278.6) that increases substantially during postnatal development (Post-pubertal TPM = 1221.2, Adult TPM = 1269.4) as the mature contractile apparatus is established.

Module 3: Metabolic Transition. This module captures genes upregulated during metabolic maturation of skeletal muscle. *MSS51* (pig FC = 3.02, human FC = 1.55) is a skeletal muscle-specific regulator of oxidative metabolism[31]. Notably, MSS51 is highly expressed specifically in fast-twitch (glycolytic) muscle fibres, with reduced expression in slow-twitch (oxidative) fibres; the increased postnatal expression reflects the maturation of fast-twitch fibre metabolism.

These findings provide quantitative evidence that pig muscle development mirrors the human program at the molecular level, validating our transcriptomic atlas as a biologically accurate reference for porcine developmental staging.

3 Discussion

We have established the first calibrated transcriptomic atlas for porcine development, providing a rigorous molecular alternative to subjective morphological staging[6]. By harmonising data across five tissues into a unified predictive framework, we demonstrate that biological maturity can be inferred with high precision solely from gene expression profiles. This shift from qualitative observation to quantitative measurement addresses a long-standing challenge in pig research: the standardisation of developmental staging across experimental cohorts and research groups.

The practical implications for pig research are substantial. Our framework enables researchers to objectively classify animals by biological maturity rather than chronological age, reducing confounding variability in studies of nutrition, growth physiology, and disease susceptibility. For example, when evaluating dietary interventions or growth-promoting treatments, synchronising cohorts by molecular developmental stage rather than age should improve statistical power and reproducibility. The tissue-specific nature of our models also allows researchers to select staging approaches appropriate to their target organ system.

Our staging framework has immediate applications for disease modelling, particularly for age-dependent muscular disorders such as Duchenne Muscular Dystrophy (DMD). Since the molecular "blueprint" for muscle development is highly conserved between pigs and humans, porcine models can provide superior physiological mirroring compared to rodents. DMD typically manifests in humans between 2–5 years of age, with rapid deterioration through the pre-teen years. Using our transcriptomic atlas, this developmental window corresponds approximately to Infant stage (0–20 d) in pigs for early onset, with pathological changes becoming visible by Pre-pubertal stage (60 d) and rapid changes occurring within Post-pubertal stage (150 d). By providing a standardised molecular definition of developmental stages, the atlas enables researchers to identify specific developmentally conserved checkpoints that a successful gene therapy must restore. For example, researchers could track whether a gene therapy successfully stabilises *PDLIM3* (Z-disc reinforcement) and *FGFRL1* (fusion control) expression back to healthy Infant or Early childhood levels. Critically, by using molecular markers to synchronise experimental cohorts rather than relying on chronological age alone, it may be possible to reduce variability in disease progression and treatment response, improving pre-clinical trial quality.

An important biological finding is the profound tissue-specificity of developmental markers. The extremely low cross-tissue feature overlap (0.3% Jaccard similarity) reflects genuine biological differences: each tissue uses distinct molecular programs for developmental progression, as evidenced by clear tissue clustering in expression space (Supplementary Fig. S2a). Within-tissue feature stability was moderate (18–60% of features stable across random seeds), which we interpret as biological gene redundancy, where multiple valid biomarker combinations exist within each tissue. This redundancy suggests robust developmental programs that can be captured by different gene sets.

To validate the biological accuracy of our atlas, we performed cross-species comparison with human skeletal muscle, revealing strong evolutionary conservation of developmental programs. Our analysis identified a core set of shared regulatory modules (including *SORCS2* and *FGFRL1*) that orchestrate tissue maturation in both species. This conservation confirms that our porcine atlas captures fundamental developmental biology rather than species-specific artefacts. Recent work on species-specific developmental tempo provides context for understanding the molecular basis of these conserved programs[32]. However, we emphasise that cross-species validation is currently limited to muscle tissue; extending this analysis to brain, liver, lung, and blood represents a critical future direction.

We acknowledge the paradox of using chronologically defined bins to train a model intended to capture biological age. In the absence of a "gold standard" for physiological maturity, chronological age serves as a noisy proxy. Machine learning theory suggests that training on large datasets with noisy labels allows the model to learn the consensus signal—here, the molecular definition of a developmental stage—while filtering out individual noise[33]. Consequently, discrepancies between model prediction and chronological age likely reflect true biological inter-individual variability (developmental acceleration or delay) rather than model error. The fact that the learned molecular signatures (*e.g.*, *SORCS2*, *FGFRL1*) tracks development in humans, despite vastly different chronological lifespans, strongly supports the conclusion that the model has captured conserved biological programs rather than mere chronological associations.

Several limitations must be acknowledged. First, our cross-species validation relies on a small human dataset (4 infants, 7 adults) with a substantial age gap. Second, the correlation analysis required parameter optimisation, though sensitivity analysis confirms robustness. Third, class imbalance necessitated reduced classification schemes for some tissues. Fourth, the reliance on bulk RNA-seq averages signals across cell types. Future iterations incorporating single-cell resolution[34,35] will allow us to dissect precisely which cell populations drive these developmental clocks.

In conclusion, this work provides a transparent, reproducible molecular standard for developmental staging in pig research. The atlas offers an immediately applicable tool for standardising experimental cohorts in swine nutrition, production, and biomedical studies. As the pig continues to serve as a major species for both agricultural and biomedical research, this resource ensures that developmental age is defined with the precision required for rigorous, reproducible science.

4 Methods

4.1 Data acquisition and processing

Bulk RNA-seq transcript-per-million (TPM) matrices and metadata were obtained from the PigGTEx resource[15,16]. Ages recorded in days were harmonised to developmental stages: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult ($>$365 d). Preprocessing included a $\log_2(x+1)$ transformation followed by variance filtering to retain genes above the 20th percentile per tissue. Gene expression values were standardised into z-scores to ensure comparability across tissues and experiments.

4.2 Machine learning framework

We implemented a reduction-based ordinal classification framework using Light Gradient Boosting Machine (LightGBM)[36]. Given the ordinal nature of developmental stages ($y \in \{0, \ldots, K-1\}$), we decomposed the K-class problem into $K-1$ binary classification subtasks (Frank & Hall method[37]), where the k-th subtask predicts the probability $\Pr(y > k \mid x)$. The final probability for each class c was derived as:

$$\Pr(y = c) = \Pr(y > c-1) - \Pr(y > c) \tag{1}$$

where $\Pr(y > -1) = 1$ and $\Pr(y > K-1) = 0$.

For feature selection, a preliminary LightGBM model was trained on the training set (70% of data), and features were ranked by total information gain across all trees. The top $d \leq 2000$ features were selected for the final model. Hyperparameters (num_leaves = 31, learning_rate = 0.05, feature_fraction = 0.8, n_estimators = 200, max_depth = 6) were chosen to balance model complexity and generalisation, with early stopping (patience = 10) to prevent overfitting.

Models were evaluated using a stratified train/test split (70%/30%). Primary performance metrics included Balanced Accuracy (BA) and Macro-F1 score (Supplementary Fig. S3; Supplementary Tables S1 and S2):

$$\mathrm{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c} \tag{2}$$

Confidence intervals (95%) were calculated using bootstrap resampling (1,000 iterations) on the test set predictions.

4.3 Cross-species evolutionary analysis

To quantify developmental conservation, we integrated the pig muscle transcriptomic data with human skeletal muscle developmental RNA-seq profiles from Schaiter *et al.*[23]. The human dataset comprised 4 infant samples (ages 7–28 months) and 7 adult samples (ages 30–56 years), all from quadriceps femoris muscle. We computed the $\log_2$ fold change of gene expression between Infant and Adult stages for both species.

Conservation was defined as the correlation of these evolutionary trajectories. We applied an adaptive thresholding approach, testing p-value thresholds from 0.05 to 0.20 and gene numbers from 15 to 60, selecting the configuration maximising $r \times \sqrt{n}$ while ensuring $r > 0.6$. The optimal set contained 36 genes ($p < 0.10$ threshold). We reported the Pearson correlation coefficient with 95% bootstrap confidence intervals (1,000 iterations).

4.4 Functional enrichment

Gene ontology (GO) enrichment of stage-informative markers was performed using g:Profiler[38], applying a Benjamini–Hochberg FDR correction ($p_{\mathrm{adj}} < 0.05$). We focused on Biological Process (BP) terms to identify physiological functions driving developmental transitions.

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Data availability

The PigGTEx data used in this study is available at `https://piggtex.farmgtex.org/`. Raw and processed RNA-seq data are available in the NCBI Gene Expression Omnibus (GEO) under accession number GSE257558.

Code availability

All code required to reproduce the findings are available at `https://github.com/TianYuan-Liu/pig-dev-stage`. All analyses were conducted using Python 3.9 and R 4.2.0.

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Author contributions

T.L. conceived the study, designed the methodology, performed all computational analyses, interpreted results, and wrote the manuscript. R.L. and I.M.K. contributed to manuscript editing and review. P.T. supervised the study, provided guidance, and edited the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information accompanies this paper.

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Figures

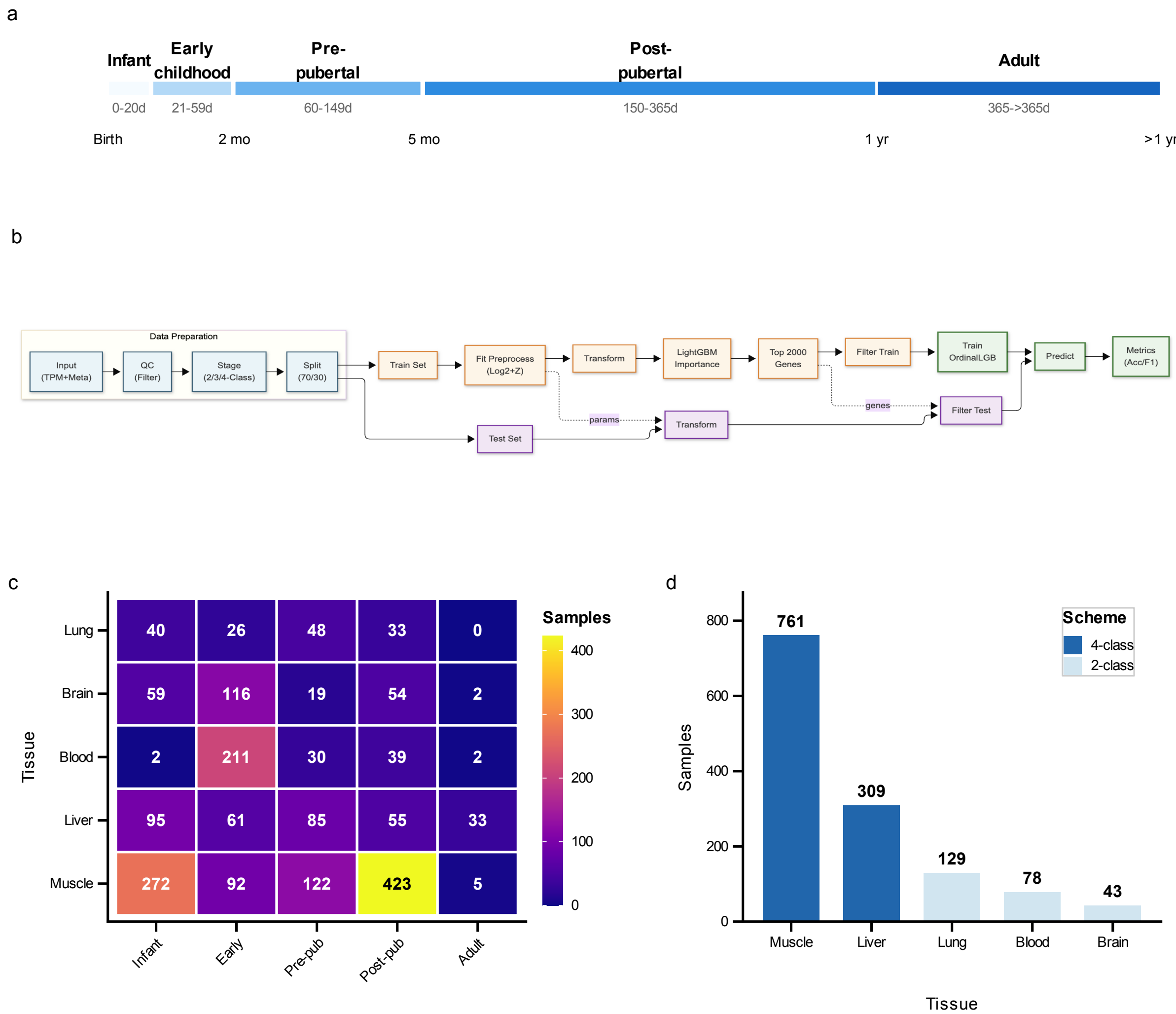

Figure 1: Study design and data landscape. a, Developmental stage definitions mapping chronological porcine age to five ordered stages: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d). **b**, Machine learning workflow diagram illustrating the pipeline from data preprocessing to model calibration and evaluation. **c**, Sample distribution heatmap showing tissue-by-stage availability across 1,924 samples from five tissues. **d**, Classification scheme assignments based on per-class sample availability, with 4-class schemes for muscle and liver, and 2-class schemes for blood, brain, and lung.

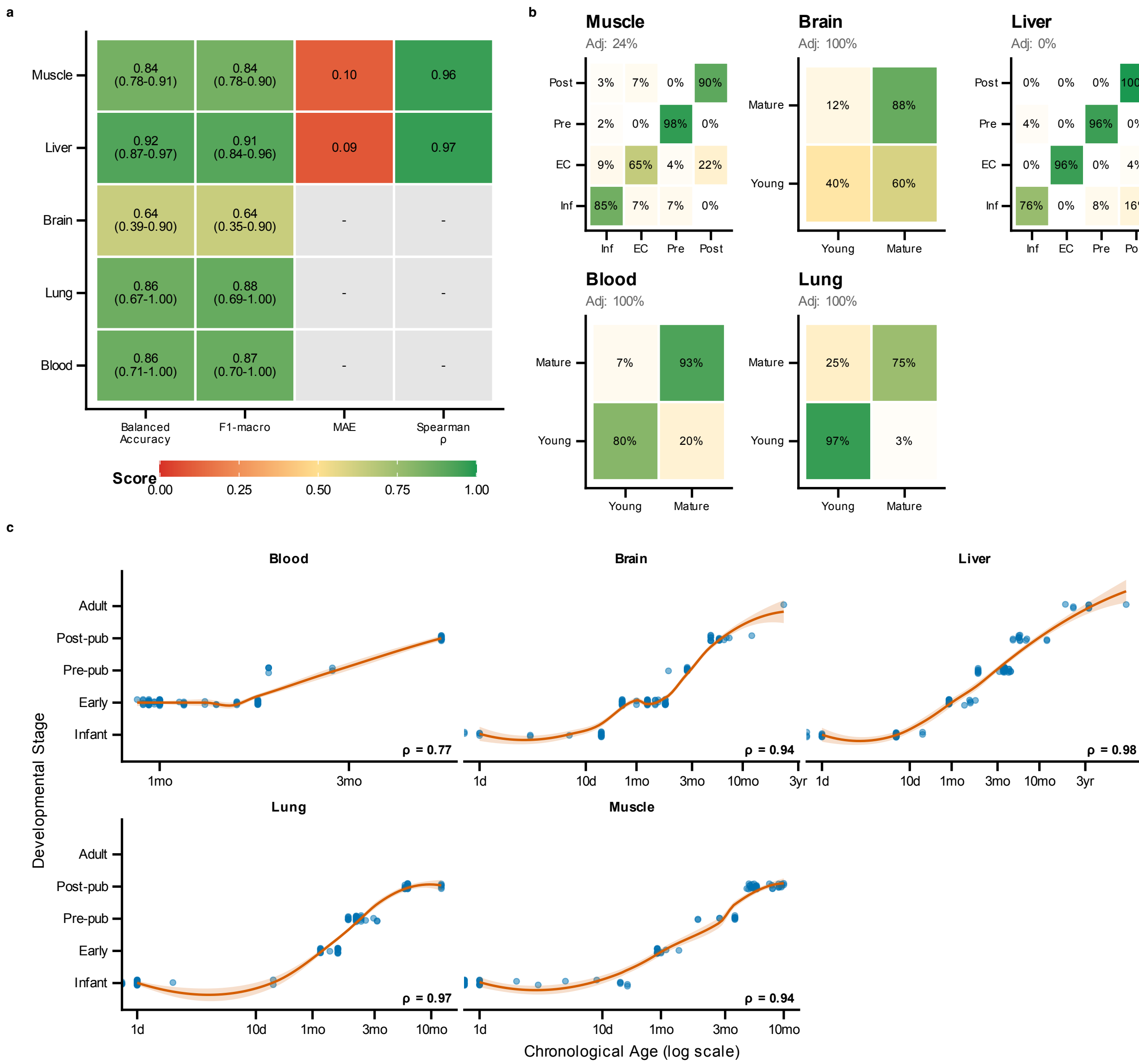

Figure 2: Classifier performance diagnostics. **a**, Performance metrics heatmap showing balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap confidence intervals across porcine tissues. Mean balanced accuracy across all tissues is 0.825, ranging from 0.638 (brain) to 0.920 (liver). **b**, Confusion matrices for each tissue showing classification accuracy and error patterns. **c**, Scatter plots of predicted developmental stage versus chronological age, demonstrating strong age–stage correlations (liver $\rho = 0.979$, muscle $\rho = 0.937$, brain $\rho = 0.942$, blood $\rho = 0.773$).

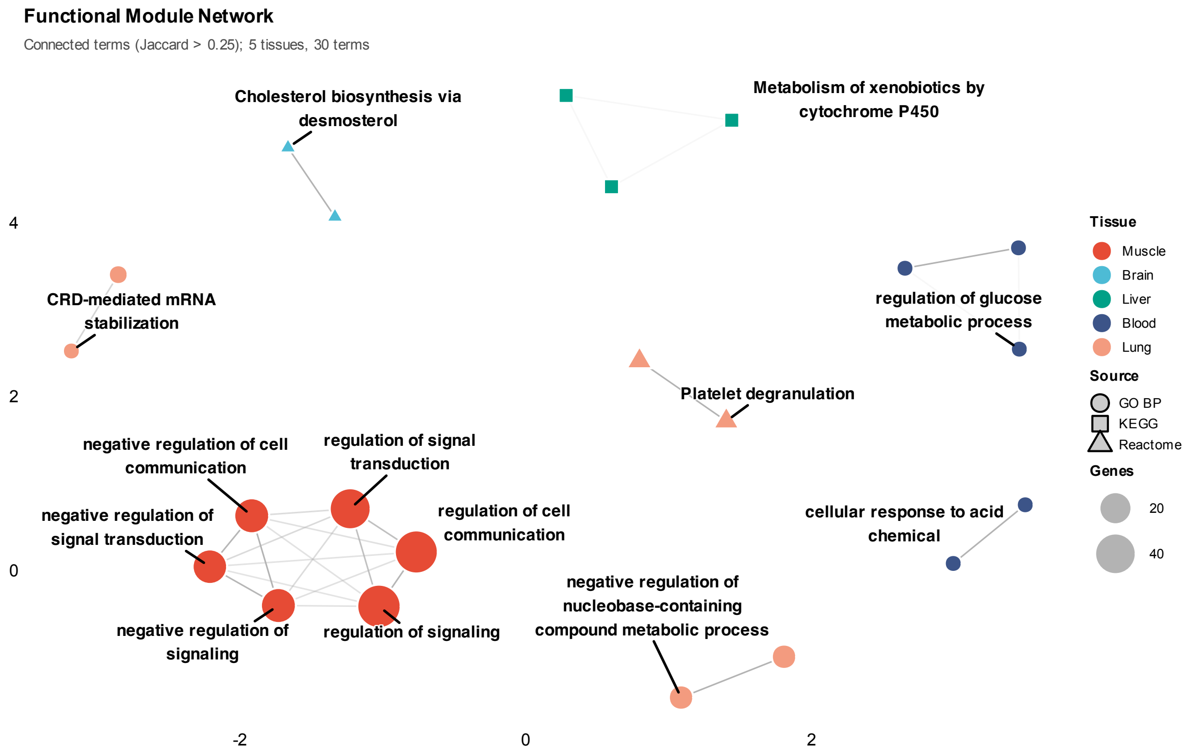

Figure 3: Functional enrichment analysis of stage-informative porcine genes. Functional network showing interconnected enrichment modules, with nodes representing enriched terms coloured by tissue and edges connecting terms sharing >25% of genes (Jaccard similarity). Hubs and cluster representatives are labelled with biological process or pathway names.

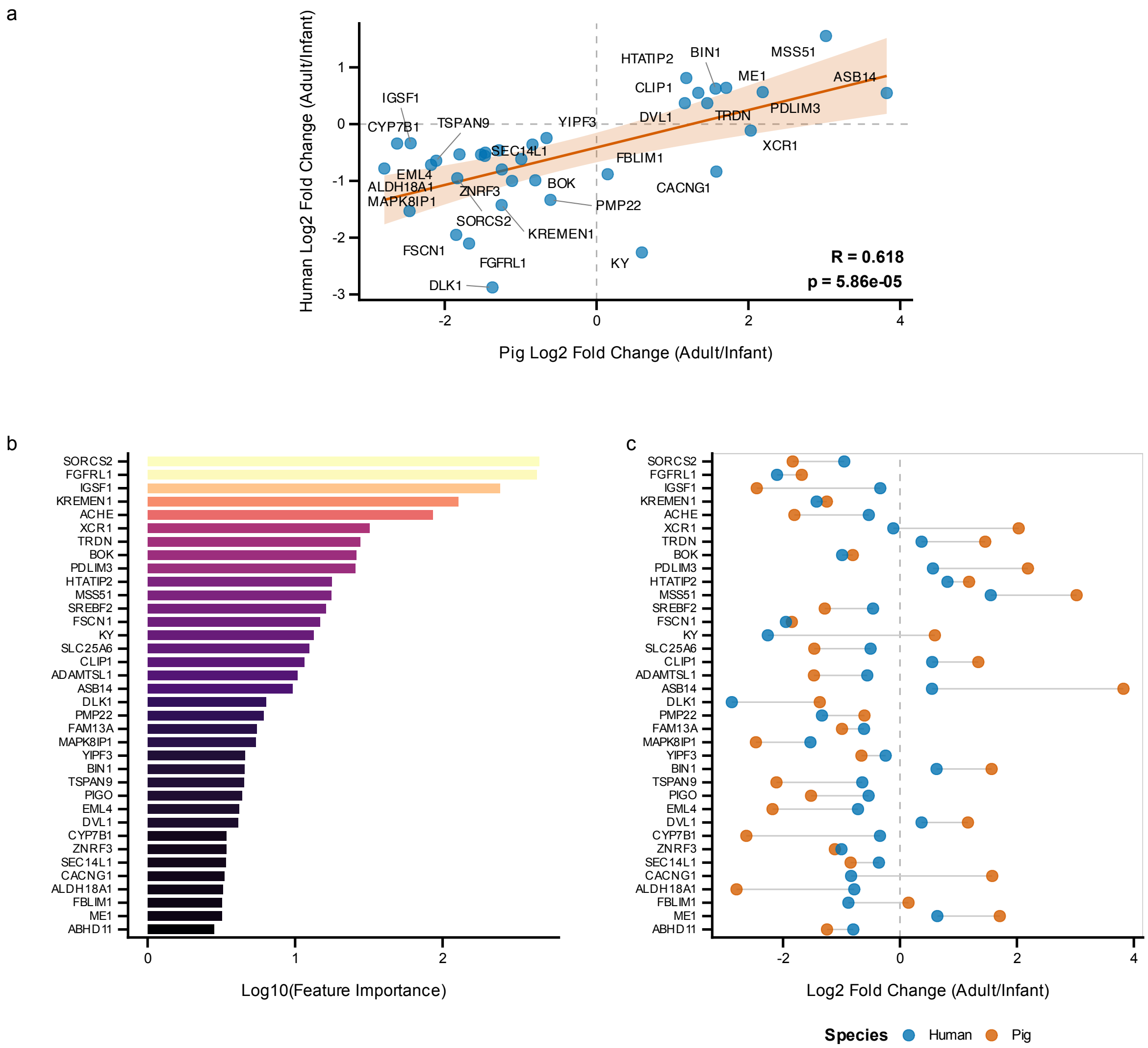

Figure 4: Cross-species validation of developmental gene expression. Human skeletal muscle transcriptomic data from Schaiter *et al.*[23] comparing infant (7–28 months) versus adult (30–56 years) samples were used to validate conservation of developmental signatures. For porcine samples, the comparison uses Infant stage (0–20 d) versus Post-pubertal/Adult stage (>150 d). The five porcine developmental stages are: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d). **a**, Scatter plot of $\log_2$ fold changes between pig and human, showing significant positive correlation (Pearson $r = 0.618$, $p = 5.86 \times 10^{-5}$, 95% CI: 0.364–0.787). Of 36 genes meeting significance criteria ($p < 0.10$) in both species, 32 (88.9%) showed conserved direction of change. **b**, Feature importance of top conserved markers in the developmental stage classifier. **c**, Paired fold change comparison showing pig and human values for each conserved gene. Top markers include *SORCS2*, *FGFRL1*, *ACTC1*, *DLK1*, *KREMEN1*, and *ACHE*.